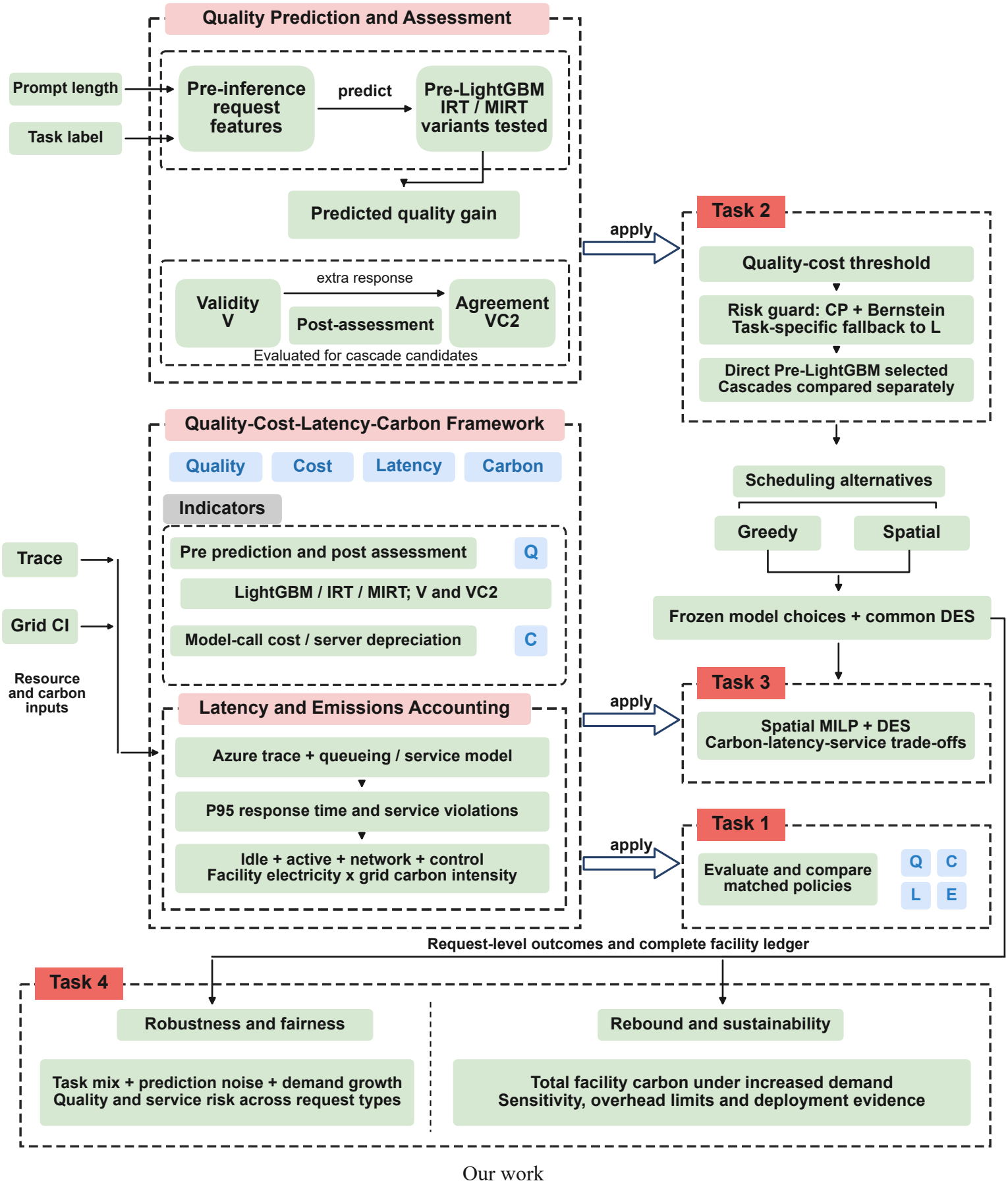

Quality Prediction and Assessment
Prompt length
Task label
Pre-inference request features
predict
Pre-LightGBM IRT / MIRT variants tested
Predicted quality gain
extra response
Validity V
Post-assessment
Agreement VC2
Evaluated for cascade candidates
apply
Task 2
Quality-cost threshold
Risk guard: CP + Bernstein Task-specific fallback to L
Direct Pre-LightGBM selected Cascades compared separately
Scheduling alternatives
Greedy
Spatial
Frozen model choices + common DES
Quality-Cost-Latency-Carbon Framework
Quality
Cost
Latency
Carbon
Indicators
Pre prediction and post assessment
Q
LightGBM / IRT / MIRT; V and VC2
Model-call cost / server depreciation
C
Trace
Grid CI
Resource and carbon inputs
Latency and Emissions Accounting
Azure trace + queueing / service model
P95 response time and service violations
Idle + active + network + control
Facility electricity x grid carbon intensity
apply
Task 3
Spatial MILP + DES
Carbon-latency-service trade-offs
apply
Task 1
Evaluate and compare matched policies
Q
C
L
E
Request-level outcomes and complete facility ledger
Task 4
Robustness and fairness
Task mix + prediction noise + demand growth
Quality and service risk across request types
Rebound and sustainability
Total facility carbon under increased demand
Sensitivity, overhead limits and deployment evidence

Our work